This policy is currently under review and the current version remains endorsed until the review is complete – September 2022

Guideline

Renal Colic

Scope:

Site	Department	Discipline
AHS	Emergency Department	All

Aim

Most cases of renal colic can be managed in and discharged from the Emergency Department. All suspected cases of renal colic should have a radiological investigation performed in the Emergency Department prior to discharge. Patients can be discharged once the pain has been controlled, and the following complications excluded:

- **an infected and obstructed system**
- **renal failure**
- **solitary kidney**

ALL patients should be referred to the Stone Clinic at RPH - fax your medical notes and Stone Clinic Referral (AKMR 1.8) to 9224 2868 or urgent ones to 6477 5199. Please ensure that patients receive a discharge advice sheet and a script for analgesia. **All complicated cases, including persistent pain, should be discussed with Urology at RPH before discharge or transfer** (see below).

Background

12% of the population make a kidney stone at some stage in their lives. The majority of kidney stones are passed spontaneously, without surgical intervention.

- 80% of stones 5mm or less in diameter will pass spontaneously.
- 50% of stones 8mm or less in diameter will pass spontaneously.
- Most stones 10 mm or more in diameter require surgical removal.

Analgesia

NSAIDs

Renal colic pain is caused by prostaglandin release. For this reason, drugs which inhibit prostaglandin release are the most effective in controlling renal colic pain. NSAIDs have been widely studied in controlling the pain from renal colic and should be considered first line treatment. These patients often have accompanying nausea and vomiting, so parenteral or rectal routes are preferred. Options include indomethacin suppositories, IV/IM paracoxib or ketorolac. Administration of an indomethacin suppository during admission to ED can demonstrate to the patient the ease and effectiveness of use following discharge.

Narcotics

Patients may also require narcotic medication to control the pain, with research indicating dual therapy is superior to either alone for pain control. Patients on ACE inhibitors in combination with diuretics are at risk of developing renal failure ("triple whammy"), and alternative pain relief should be used in these patients. Pregnant patients should not be given NSAIDs.

Medical Expulsion Therapy

There has been considerable debate about the usefulness of alpha blockers to speed the passage of stones through the ureter. The most recent meta-analysis of the literature supports the use of alpha blockers for patients with stone sizes of 5-10mm. Tamsulosin (400mcg daily) improved time to stone passage, improved rates of passage, improved pain levels and was well tolerated.

Imaging for renal colic

The diagnosis of renal colic should be confirmed with appropriate radiological imaging whilst minimising radiation exposure.

First stone presentation

The most sensitive test for urolithiasis is a non-contrast CT scan using stone protocol. A CT KUB is performed as a "low dose" CT scan, which is adequate for diagnosis with minimal radiation. If other diagnoses are being considered, it is better to indicate the differential clinical diagnosis on the request form, and the Radiologist will decide on whether a pre- and post-contrast CT scan is indicated. Included on the request form should be a request for reporting on the density of the stone (Hounsfield units), as this assists the choice of follow-up imaging by Urology. Ensure that you request PRC put the imaging on PACS so that the Urology team at RPH can access it at the time of follow-up.

Re - presentation for an existing stone

If a stone has been documented on CT scan for this episode, repeating the CT scan when a patient comes back with a stone related problem usually not necessary. KUB Xray and USS are usually adequate for planning management.

Often decisions can be made without imaging. For example, if the patient has come back for pain not controlled with NSAID suppositories, they require surgical intervention on a clinical basis, and repeat imaging is unnecessary. Similarly, if the patient returns infected obstructed, they need emergency decompression, not another CT scan! The urology team should be contacted in these circumstances, and they will organise an inpatient procedure for the patient.

Recurrent stone former

For those patients with a well-documented history of recurrent stone formation and renal colic, the aim is for diagnosis with minimal radiation exposure. The preference is for KUB Xray and renal USS.

Repeat CT scanning should be avoided, when possible. If the stone density on previous CTKUB is <600 HU, ureteric calculi are unlikely to be seen on plain XR. They may also be obscured because they overlie the sacrum, or are confused with phleboliths.

Pregnancy

Pregnant patients should have ultrasound only initially. Occasionally the Urology team may request an MRI pyelogram. All pregnant patients should be assessed clinically by urology. As mentioned above, they should not receive NSAIDs.

Other investigations

These tests should be done routinely:

- MSU if suspect UTI
- U&E / Creatinine
- Serum calcium and uric acid levels

Complications

The following patients **must** be discussed with the Urology Registrar:

Infected obstructed kidney

- Febrile patient, fever >/= 38º C, often with a history of rigors.
- Imaging evidence of obstruction, usually with a stone visible in the ureter or pelvi-ureteric junction. Hydronephrosis is usually present (rarely no hydronephrosis

This is **a urological emergency**.

These patients should have blood and urine cultures taken, followed by administration of intravenous antibiotics, usually gentamicin and amoxicillin (as per eTG). These patients require urgent admission and intervention with either cystoscopy and JJ stent insertion or percutaneous nephrostomy drainage to unblock the collecting system. The urology team will arrange this.

Renal failure or solitary kidney

Generally, a creatinine level > 200 micromol/L is an indication to unblock a kidney. Commonly these patients have a known diagnosis of impaired renal function prior to their renal colic.

Patients with renal colic will often have a mild elevation in serum creatinine, in the order of 140 – 160 micromol/L. There is no rational nephrological explanation for this. It always returns to baseline within a day or two, even if the stone hasn't passed.

Patients with a solitary kidney require urgent intervention to unblock their kidney, either cystoscopy and JJ stent, or percutaneous nephrostomy tube placement by radiology. The urology team will arrange a plan for intervention.

Systemic complications

Patients may require urgent management of associated problems such as hyperkalaemia or uraemic symptoms. Again, this should be discussed with the urology team. The assistance of the nephrologists may be requested in this situation.

Persistent pain

Patients with pain that is persisting or frequently recurrent and thus causing significant morbidity should be discussed with the Urology Registrar and considered for admission for early intervention.

Stones > 6mm

Stones > 6mm are unlikely to pass without intervention.

Does this patient need urgent intervention?

- **Hydronephrosis** seen on CT scan
  This is normal with a ureteric stone and is not an indication to intervene. It reflects ureteric obstruction which is normal and usually intermittent as a stone is passing. Persisting high grade obstruction for more than a couple of weeks can damage kidneys permanently. This fortunately is rare. Historically, no major renal damage has been seen when patients are reviewed in a timely manner. Significant persisting hydronephrosis at 6 weeks is an indication to unblock the kidney. Patients that develop significant permanent renal damage (usually at one to two years after an identifiable episode of renal colic) have not been diagnosed and followed up correctly i.e. nobody ensured that the stone from that episode had passed. Follow-up should occur within three months (some suggest one month and some suggest six weeks – there is no convincing evidence for a shorter interval) of the original episode – either with imaging or with the stone seen in a jar.

- **A positive MSU** test
  Not uncommon in women presenting with renal colic, but does it **not** mean the patient is "infected obstructed". Infected obstructed is a clinical diagnosis, with a patient clinically septicaemic and a temperature recorded >38 degrees C. If the patient is not infected obstructed or septicaemic clinically, the patient is considered to have two pathologies i.e. renal colic – managed by the rules outlined above, plus a lower urinary tract infection (or asymptomatic bacteriuria), which should be managed as a lower urinary tract infection with oral antibiotics. If discharged, it must be emphasised to the patient that they should return if they become unwell with rigors and a temperature >38 degrees C. Note that only 30% of patients with an infected obstructed kidney have a positive MSU.

- **Raised WBC count**
  Normal in acute renal colic, with no bearing on diagnosis of infected obstructed kidney.

Related Policy and Procedure Documents

RPH Renal Colic Protocol

Related Standards

- NSQHS Standard: 1.1 Governance for safety and quality in health service organisation
- EQuIP Standard 12: Provision of Care

References

RPH Renal Colic Protocol

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Authorisation

Authorising Committee	Date Authorised	Date of Next Review
Emergency Department Safety and Quality Meeting	17 4 2018	17/04/2021
Director Clinical Services	09/09/2022	09/09/2023

Appendix 1: Discharge advice for patient’s with Renal Colic

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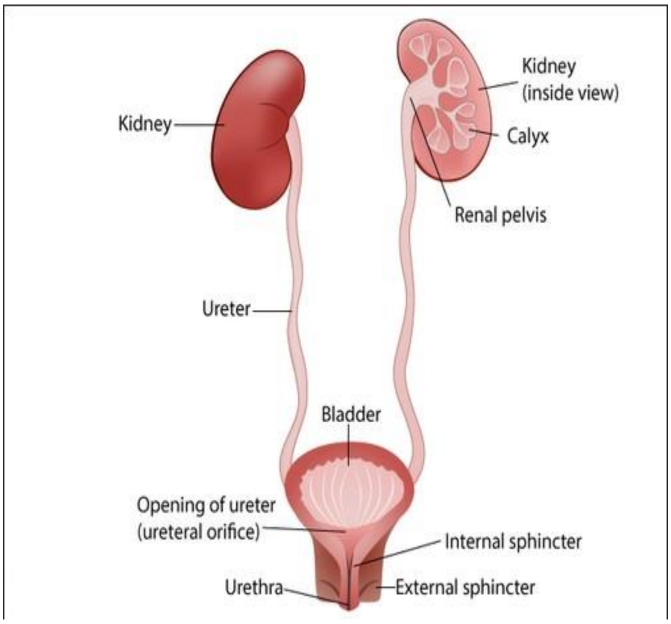
Your doctor has diagnosed your condition as renal colic (kidney stone). The investigations have shown that you have a calculus (stone) in your ureter (tube connecting kidney with bladder).

12% of the population make a kidney stone at some stage of their lives. Most stones (80%) will pass spontaneously; that is they will travel from the kidney, down the ureter and pass into the bladder, where they are no longer of concern. The probability that a stone will pass into the bladder is related to the size and position of the stone in the ureter. As intuition would suggest, larger stones and stones high up in the ureter are less likely to pass than smaller stones that have travelled to the bottom of the ureter.

In one study, it was found that the chance of a stone passing spontaneously within 20 weeks, based on size alone was:

0-3 mm	98%
4 mm	81%
5 mm	65%
6 mm	33%
> 6.5 mm	9%



The size of your stone is ___ mm and is positioned ________________

As the stone moves down the ureter, it may cause you more pain. A stone in the lower third of the ureter may cause some bladder irritation and result in symptoms of "urgency" (sensation of urgently wanting to pass urine) and "frequency" (sensation of needing to pass urine frequently). Once the stone passes into the bladder, these symptoms subside. Cessation of pain does not mean that you have necessarily passed the stone.

The most effective drugs in controlling episodes of renal colic pain are anti-inflammatories (NSAIDs); they are considered first line treatment. As nausea and vomiting often accompany episodes of severe renal colic pain, the rectal route of administration is preferred. For this reason, your doctor may prescribe an anti-inflammatory suppository such as indomethacin. You should NOT take anti-inflammatories if you are pregnant, have asthma which is worsened by anti-inflammatories, have peptic ulcer disease or kidney function impairment, or are taking a combination of antihypertensive medications (ACE- inhibitors + diuretic).

We recommend that you return to all your normal daily activities. You may wish to carry some NSAID suppositories with you in case you get an episode at work.

Paradoxically, increasing your oral fluid intake slows the passage of the stone and it recommended that you drink normally while passing the stone. Once the stone has passed, you can halve your chances of making another stone if you can maintain a urine output of 2 litres per 24 hours in the long term.

If you pass the stone and are able to retrieve it – best done by straining all urine through a sieve – please take it to your clinic appointment for biochemical analysis.

In our department we usually refer all public patients with renal colic to a public outpatient Urology clinic at Royal Perth Hospital. You will be sent an appointment in the mail. This follow-up is to ensure that the stone passes and that no damage to the kidney occurs due to blockage of the flow of urine from the kidney. Surgical or radiological treatment is sometimes necessary to remove the stone from the ureter. The clinic may also provide advice regarding prevention of further stone formation. Please attend your appointment, even if the renal colic pain has not recurred, as it does not necessarily mean that you have passed the stone. If you do not receive an appointment, please see your GP for re-referral to the Stone Clinic.

The urgency of follow-up depends on various clinical factors and ranges from immediate to semi-urgent (usually within 2 or 6 weeks, but may be up to 3 months).

To expedite follow-up, you may choose to see an Urologist privately. It is likely that you will be charged a fee and the fee varies according to your circumstances (privately insured, pensioner, public patient).

Please return to the Emergency Department immediately if you:

- develop a fever greater than 38 degrees
- experience pain that is not controlled with NSAIDs

References:

1. Jendeberg, J., Geijer, H., Alshamari, M. et al. Eur Radiol (2017) 27: 4775.